```
//Aufgabe 1 ------------------------
int input1 = 5;
int input2 = 5;

bool isEqual = input1 == input2;


//Aufgabe 2 ------------------------
int input = 15;
bool isEven = input % 2 == 0;

if (isEven)
{
    Console.WriteLine("ist gerade");
}
else
{
    Console.WriteLine("ist ungerade");
}


//Aufgabe 3 ------------------------
int number = 15;
bool isPositive = number > 0;

if (isPositive)
{
    Console.WriteLine("ist positiv!");
}
else if (number == 0)
{
    Console.WriteLine("ist null!");
}
else
{
    Console.WriteLine("ist negativ!");
}


//Aufgabe 4 ------------------------
int year = 2016;
bool isLeapYear = (year % 4 == 0) && (!(year % 100 == 0) || year % 400 == 0);

if (isLeapYear)
{
    Console.WriteLine($"Year: {year} is a leap year!");
}
else
{
    Console.WriteLine($"Year: {year} is no Leap year!");
}


//Aufgabe 5 ------------------------
int number1 = 25;
int number2 = 64;
int number3 = 10;

int greatestNumber;

if (number1 > number2)
{
    if (number1 > number3)
    {
        greatestNumber = number1;
    }
    else
    {
        greatestNumber = number3;
    }
}
else
{
    if (number2 > number3)
    {
        greatestNumber = number2;
    }
    else
    {
        greatestNumber = number3;
    }
}

Console.WriteLine($"The greatest number is {greatestNumber}");


//Aufgabe 6 ------------------------
int x = 7;
int y = 9;

string quadrant = "";

if (x > 0 && y > 0)
{
    quadrant = "1nd Quandrant";
}
else if (x < 0 && y > 0)
{
    quadrant = "2nd Quandrant";
}
else if (x < 0 && y < 0)
{
    quadrant = "3nd Quandrant";
}
else if (x > 0 && y < 0)
{
    quadrant = "4nd Quandrant";
}
else
{
    quadrant = "no Quadrant";
}

Console.WriteLine($"The number is in {quadrant}.");


//Aufgabe 7 ------------------------
int side1 = 50;
int side2 = 50;
int side3 = 60;

if (side1 == side2 || side2 == side3 || side3 == side1)
{
    if (side1 == side2 && side2 == side3)
    {
        Console.WriteLine($"Das Dreieck ist gleichseitig.");
    }
    else
    {
        Console.WriteLine($"Das Dreieck ist gleichschenklig.");
    }
}
else
{
    Console.WriteLine($"Das Dreieck hat unterschiedliche Seitenlängen.");
}



//Aufgabe 8 ------------------------
int menuSelection = 1;
int userInput = 5;
int userInput2 = 5;
double result = 0;

if (menuSelection == 1)
{
    result = userInput * Math.PI * Math.PI;
}
else if (menuSelection == 2)
{
    result = userInput * userInput2;
}
else
{
    result = userInput * userInput2 / 2;
}

Console.WriteLine("Ergebnis 8: " + result);

//Aufgabe 9 ------------------------
int menuSelection2 = 1;
int firstInput = 10;
int secondInput = 2;
double result2 = 0;

if (menuSelection2 == 1)
{
    result2 = firstInput + secondInput;
}
else if (menuSelection2 == 2)
{
    result2 = firstInput - secondInput;
}
else if (menuSelection2 == 3)
{
    result2 = firstInput * secondInput;
}
else if (menuSelection2 == 4)
{
    result2 = firstInput / secondInput;
}
else if (menuSelection2 == 5)
{
    result2 = firstInput % secondInput;
}

Console.WriteLine("Ergebnis 9: " + result2);



//Alternative Lösungen für 7 + 8 ---------------------------------------------------

//Alternative Aufgabe 8 ------------------------
int menuSelectionAlt = 1;
int inputAlt = 5;
int inputAlt2 = 5;
double resultAlt = 0;

switch (menuSelectionAlt)
{
    case 1:
        resultAlt = inputAlt * Math.PI * Math.PI;
        break;
    case 2:
        resultAlt = inputAlt * inputAlt2;
        break;
    default:
        resultAlt = inputAlt * inputAlt2 / 2;
        break;
}

Console.WriteLine("Ergebnis 8 alt: " + resultAlt);


//Alternative Aufgabe 9 ------------------------
int menuSelectionAlt2 = 1;
int firstInputAlt = 5;
int secondInputAlt = 5;
double resultAlt2 = 0;

switch (menuSelectionAlt2)
{
    case 1:
        resultAlt2 = firstInputAlt + secondInputAlt;
        break;
    case 2:
        resultAlt2 = firstInputAlt - secondInputAlt;
        break;
    case 3:
        resultAlt2 = firstInputAlt * secondInputAlt;
        break;
    case 4:
        resultAlt2 = firstInputAlt / secondInputAlt;
        break;
    default:
        resultAlt2 = firstInputAlt / secondInputAlt;
        break;
}

Console.WriteLine("Ergebnis 8 alt: " + resultAlt2);
```